

Practice Quiz 06D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 06

Existence and uniqueness as two separate claims.

Label and prove both halves.

1. Prove that for every real number $a \neq 0$, there is exactly one real number x such that $2ax + a = 5a$.

Solution Let $a \in \mathbf{R}$ and suppose $a \neq 0$.

Existence. Set $x = 2$. Then

$$2ax + a = 2a(2) + a = 5a,$$

so a solution exists.

Uniqueness. Suppose x and y both satisfy the equation. Then $2ax + a = 5a = 2ay + a$. Subtracting gives $2a(x - y) = 0$. Since $2a \neq 0$, we have $x = y$. Thus the solution is unique. ■